MMGCF: Multimodal Graph Collaborative Filtering for Recommendation with Graph Convolutional Networks

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Abstract

The growing interest of the research community in multimodal recommendation models has introduced significant challenges, most notably how to effectively process and integrate multimodal features into traditional approaches such as collaborative filtering. Despite the efforts, many existing methods achieve only marginal performance gains at the cost of increased model complexity and longer training time. In this paper, we present MMGCF (**M**ulti**M**odal **G**raph **C**ollaborative **F**iltering), a recommendation model that integrates an efficient collaborative filtering backbone with pre-trained multimodal item features, thereby combining the advantages of multimodal representations with negligible additional model complexity. Through extensive experiments, we evaluate MMGCF against state-of-the-art recommendation models. The results demonstrate that our model achieves significant performance gains with minimum overhead, maintaining competitive training times. Our source code is available at https://github.com/swapUniba/MMGCF

CCS Concepts

• **Information systems** → **Recommender systems**.

Keywords

Recommender Systems, Multimodal, Graph Convolutional Networks

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1 Introduction and Motivations

In recent years, the research community has shown growing interest in multimodal deep learning [14], spanning multiple domains, from natural language processing [10] to computer vision [1]. The field of Recommender Systems (RS) is no exception, and an expanding body of literature now explores how to effectively leverage multimodal data to enhance recommendation accuracy and personalization [11, 20, 26]. Multimodal RSs typically exploit multimodal features associated with items, such as textual descriptions, images, movie posters, or video trailers, to enrich item representations. By incorporating these different signals, RSs generate more expressive embeddings that significantly enhance recommendation accuracy [18, 30]. However, a key challenge remains: the effective integration of these multimodal features with traditional ID embeddings [7], that are learned through traditional Collaborative Filtering (CF). The most common approach involves learning user and item-ID embeddings from historical interaction data, then combining them with multimodal item features through a fusion mechanism.

For example, VBPR [6] adopts the BPR loss [16] to learn ID embeddings and enhances item representations by concatenating visual features with item ID embeddings. While this approach is computationally efficient, it often provides marginal performance gains, as its simple fusion strategy fails to capture the latent correlations between disparate feature spaces. More sophisticated approaches exploit graph structures, which already have shown their effectiveness in RSs [13, 15]. MMGCN [25], based on Graph Convolutional Networks (GCNs) [29], integrates multimodal data into a graph-based CF framework. This is achieved by constructing modality-aware item representations that are subsequently propagated across the graph via convolutional layers. However, this design introduces additional parameters and computational overhead due to modality-specific convolutions. GRCN [24] evolves the graph-based paradigm by encoding both local and global contextual dependencies through relational graph convolutions. This approach facilitates the simultaneous propagation of CF signals and multimodal features. However, this integration comes at the expense of increased model complexity. Similarly, LATTICE [28] leverages multimodal features to discover latent structures within item representations. By explicitly constructing a latent item-item graph, the model captures high-order dependencies which are which are then aggregated using GCNs. FREEDOM [32] adopts a decoupled multimodal learning strategy, coordinating CF and multimodal representation learning via modality-specific encoders and fusion mechanisms, achieving strong performance with a more complex optimization pipeline. Overall, existing multimodal RSs often rely

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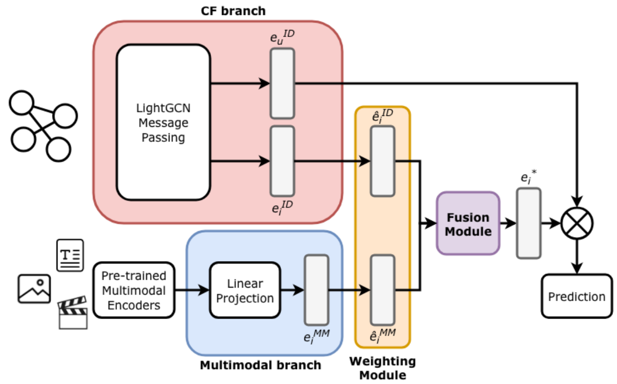

Figure 1: Workflow of MMGCF model.

on highly sophisticated architectures to integrate multimodal signals. However, the resulting performance gains are often marginalized by a substantial increase in computational overhead. This trade-off underscores a critical gap in the field: the need for efficient yet robust architectural solutions. To this end, we propose MMGCF (MultiModal Graph Collaborative Filtering), a multimodal RS built upon the high-efficiency LightGCN backbone [7]. MMGCF prioritizes architectural simplicity by integrating multimodal features through precise, low-impact strategies.

Specifically, the proposed model augments LightGCN's item ID embeddings with multimodal signals using lightweight weighting and fusion mechanisms. This design avoids complex multimodal graph architectures, while still achieving strong performance with low computational overhead.

The main contributions of this work are summarized as follows:

- *Efficient Architecture:* MMGCF enriches a robust CF backbone with modality-specific features, achieving more precise item representations without the computational toll of traditional graph-based integration.
- *Agnostic Fusion Strategies:* We systematically investigate various weighting and fusion strategies to effectively fuse CF signals with multimodal data. These strategies provide versatile design configurations while maintaining minimal computational overhead.
- *Empirical Validation:* Through extensive benchmarking across multiple datasets, we show that MMGCF consistently achieves competitive or better performance than current State-of-the-Art (SOTA) multimodal models, while maintaining efficient training times.

2 MMGCF Architecture

The MMGCF Architecture is organized into four distinct components: *(i)* a backbone CF model based on LightGCN, which provides user and item ID embeddings; *(ii)* a multimodal branch that processes item features from multiple modalities; *(iii)* a Weighting Module that controls the relative contribution of ID and multimodal signals; *(iv)* a Fusion Layer that aggregates the weighted embeddings into a unified item representation. The resulting item embedding is then used with the user ID embedding to provide recommendations. This workflow is illustrated in Figure 1 and elaborated upon in the reminder of this section.

CF branch. This branch captures CF signals from the interaction matrix $\mathbf{R} \in \mathbb{R}^{M \times N}$, with $\mathbf{R}_{i,j} = 1$ denoting an observed positive interaction between user i and item j, and $\mathbf{R}_{i,j} = 0$ otherwise. We adopt LightGCN [7] as our backbone model to learn d-dimensional user and item ID embeddings. It is chosen for its proven effectiveness and efficiency. The model performs linear message passing across K layers of the bipartite user-item graph. The final ID embeddings, $\mathbf{e}_u^{ID}$ and $\mathbf{e}_i^{ID}$, are then obtained by aggregating the representations across layers. For a comprehensive discussion on the propagation mechanics and architectural advantages, we refer the reader to the original work [7].

Multimodal Branch. The subsequent stage of the workflow processes multimodal item embeddings. Each set of embeddings coming from the M multimodal data sources (e.g., images, text, video) is projected through a learnable linear layer into the same latent space as the ID embeddings. This projection aligns the heterogeneous feature spaces of different modalities with the collaborative embedding space.

Formally, let $\mathbf{i}_m$ be the item embedding derived from modality m. The linear projection is defined as:

$$\mathbf{e}_i^m = \mathbf{W}_m \mathbf{i}_m + \mathbf{b}_m \tag{1}$$

where $\mathbf{W}_m \in \mathbb{R}^{d \times d_m}$ and $\mathbf{b}_m \in \mathbb{R}^d$ are the learnable weight matrix and bias vector, respectively, mapping the d_m-dimensional modality features to the d-dimensional collaborative embedding space.

Weighting Module. Once item ID and multimodal embeddings are processed, we apply a weighting mechanism to balance the ID embeddings $\mathbf{e}_i^{ID}$ and the projected multimodal embeddings $\mathbf{e}_i^m$. We propose three distinct strategies to be evaluated in our experiments:

- **Equal Weighting:** Both signals are treated with uniform importance, assuming a balanced contribution from both sources. This strategy serves as a parameter-free baseline, integrating the embeddings without additional scaling:

$$\hat{\mathbf{e}}_i^{ID} = \mathbf{e}_i^{ID}, \quad \hat{\mathbf{e}}_i^m = \mathbf{e}_i^m \tag{2}$$

- **Alpha Weighting:** We introduce a learnable parameter $\alpha \in [0,1]$ to dynamically regulate the relative contribution of each signal. To constrain α within the valid range of $[0,1]$, the parameter is passed through a sigmoid activation function. The weighted embeddings are defined as:

$$\hat{\mathbf{e}}_i^{ID} = \alpha \mathbf{e}_i^{ID}, \quad \hat{\mathbf{e}}_i^m = (1-\alpha)\mathbf{e}_i^m \tag{3}$$

- **Normalization:** To mitigate magnitude imbalances between the ID and multimodal signals, we apply L_2 normalization [9] to the embeddings. Since each item is associated with M modalities, we scale the ID embeddings by a factor of M to compensate for the reduction in magnitude introduced by normalization and preserve the strength of collaborative signals.

$$\hat{\mathbf{e}}_i^{ID} = M \cdot \frac{\mathbf{e}_i^{ID}}{\|\mathbf{e}_i^{ID}\|_2}, \quad \hat{\mathbf{e}}_i^m = \frac{\mathbf{e}_i^m}{\|\mathbf{e}_i^m\|_2} \tag{4}$$

This calibration ensures that both signals are optimally prepared to be fused into a unified representation.

Item Embedding Fusion. After the weighting process, the item ID and multimodal embeddings are fused to obtain a unified item representation $\mathbf{e}_i^*$. We explore three distinct strategies for this integration:

- **Mean:** the ID and multimodal embeddings are averaged over each dimension as follows:
$$\mathbf{e}_i^* = \text{avg}(\hat{\mathbf{e}}_i^{ID}, \hat{\mathbf{e}}_i^1, \ldots, \hat{\mathbf{e}}_i^M) \tag{5}$$
- **Summation:** the ID and multimodal embeddings are combined via element-wise addition:
$$\mathbf{e}_i^* = \hat{\mathbf{e}}_i^{ID} + \hat{\mathbf{e}}_i^1 + \cdots + \hat{\mathbf{e}}_i^M \tag{6}$$
- **Concatenation:** the ID and multimodal embeddings are concatenated and projected onto the same dimension of the ID embeddings. Let $\mathbf{W}_f$ be a learnable parameter and b_f a bias vector. Then, the concatenation fusion strategy is formalized as follows:
$$\mathbf{e}_i^* = \mathbf{W}_f[\hat{\mathbf{e}}_i^{ID} \| \ldots \| \hat{\mathbf{e}}_i^M] + \mathbf{b}_f \tag{7}$$

Ultimately, this process provides a unified item embedding e_i^*, which serves as a comprehensive representation by encoding both CF signals and rich multimodal features.

Model Optimization. To optimize MMGCF, we employ the Bayesian Personalized Ranking (BPR) loss [16], a pairwise objective function that encourages the model to assign higher prediction scores to observed (positive) interactions than to unobserved (negative) ones. For the sake of brevity, we refer the reader to the original work [16] for further details.

Model strenghts and advantages. The key features of our approach are:

- **Robust CF backbone:** MMGCF leverages LightGCN [7], a SOTA model that achieves strong performance while maintaining low complexity.
- **Adaptive weighting mechanisms:** Differently from the common approach of treating each modality with equal weight [6, 12, 25, 31], we introduce several lightweight yet effective alternatives. This approach is designed to enhance recommendation accuracy without introducing the heavy computational overhead typical of more complex fusion architectures [21, 24, 27, 28, 32].
- **Versatile Fusion Strategies:** Rather than relying solely on concatenation, a common practice in the literature [6, 12, 31], we evaluate a variety of fusion techniques to identify optimal methods for integrating heterogeneous signals.

3 Experimental Session

We evaluate our proposed methodology by addressing the following Research Questions (RQs):

- **RQ1:** How does MMGCF perform compared to SOTA baselines?
- **RQ2:** To what extent do hyper-parameters influence the overall performance of the model?

Datasets. We evaluate our approach using three benchmarks in multimodal recommendation: MovieLens-1M[1] (ML1M), DBbook and Sports. We consider the ML1M and DBbook versions presented in [18, 19], while the Sports dataset is provided by the Amazon Review dataset collection [31]. The available modalities vary by dataset: ML1M includes text (T), images (I), audio (A) and video (V), while DBbook and Sports are limited to text (T) and images (I). As multimodal encoders, we considered MiniLM [23] for text, ViT [4] for images, VGGish [8] for audio, and R(2+1)D [22] for videos. Statistics of the datasets are reported in Table 1.

Table 1: Statistics of the datasets.

Dataset	#Users	#Items	#Inter.	Sparsity	Modalities
ML1M	6,040	2,980	942,703	94.76%	T, I, A, V
DBbook	6,179	4,191	121,779	99.53%	T, I
Sports	35,598	18,357	296,337	99.95%	T, I

Evaluation Protocol. Following standard literature [27, 28] we split each dataset into train, validation and testing sets using an 80-10-10 ratios. The BPR loss samples 1 negative item at time. We assess model performance by considering Recall@k and NDCG@k, with $k = \{10, 20\}$, while we measure the computational efficiency is by considering the training time (seconds) per epoch. All models are evaluated in the *AllItems* setting [2].

Experimental Setting. We train our models for 500 epochs, with learning rate set to 0.001 and batch size set to 2048. We conduct a sensitivity analysis of MMGCF across three key hyperparameter: GCN backbone depth, weighting mechanism, and multimodal fusion. Specifically, we vary the number of GCN layers in $1, 2, 3$, evaluate *equal*, *alpha*, and *normalization* weighting mechanisms, and compare *mean*, *summation*, and *concatenation* fusion strategies.

Baselines and Implementation Details. We benchmark our approach against six SOTA models implemented in MMRec [31]: VBPR [6], MMGCN [25], GRCN [24], LATTICE [28], FREEDOM [32], and LGMRec [5]. All models are fully optimized through a grid search, and the full list of considered hyperparameters is reported in our repository. We run each parameter combination once, using the same seed. Statistical significance is assessed by using a paired t-test on the recommendation lists produced by our model and the best-performing baseline model, with a threshold of $p < 0.05$. We conduct our experiments on a machine running Ubuntu 22.04, equipped with an Intel(R) Xeon(R) Silver 4309Y CPU, and an NVIDIA A16 GPU. We develop our model in Pytorch 2.9 with CUDA drivers v12.8. The full codebase is available in our repository[2].

4 Results Discussion

To answer RQ1, Table 2 provides a performance comparison between MMGCF and the selected baselines across the three datasets. Figure 2 illustrates the training times (per epoch) of MMGCF, LightGCN, and the top-performing multimodal baseline model.

On ML1M, LightGCN emerges as the strongest baseline, suggesting that CF methods remain highly effective on dense dataset (see Table 1) compared to multimodal models. Nevertheless, MMGCF achives statistically significant gains over LightGCN in both Recall and NDCG, with only a marginal increase in training time per epoch (see Figure 2). These results suggest that MMGCF effectively leverages multimodal features to enhance recommendation performance without introducing significant computational overhead.

[1]https://grouplens.org/datasets/movielens/

[2]https://github.com/swapUniba/MMGCF

Table 2: Performance Comparison between MMGCF and the baseline models. Best overall results are reported in bold, while the top-performing baselines are underlined. Values with (*) indicates statistical significance at $p < 0.05$.

	ML1M				DBbook				Sports			
Model	R@10	N@10	R@20	N@20	R@10	N@10	R@20	N@20	R@10	N@10	R@20	N@20
LightGCN	0.1672	0.2600	0.2536	0.2678	0.1725	0.1306	0.2453	0.1558	0.0720	0.0410	0.1024	0.0489
VBPR	0.1368	0.1914	0.2156	0.2080	0.1955	0.1498	0.2679	0.1750	0.0546	0.0313	0.0803	0.0380
MMGCN	0.1465	0.2072	0.2339	0.2259	0.1006	0.0689	0.1644	0.0909	0.0345	0.0184	0.0543	0.0235
GRCN	0.1579	0.2256	0.2527	0.2454	0.1856	0.1395	0.2605	0.1653	0.0614	0.0339	0.0936	0.0422
LATTICE	0.1669	0.2340	0.2627	0.2542	0.2073	0.1577	0.2872	0.1855	0.0775	0.0436	0.1128	0.0527
FREEDOM	0.1565	0.2223	0.2494	0.2418	0.1825	0.1342	0.2607	0.1612	0.0762	0.0407	0.1176	0.0514
LGMRec	0.1653	0.2308	0.2585	0.2503	0.1771	0.1304	0.256	0.1575	0.0708	0.038	0.1046	0.0467
MMGCF	**0.1871***	**0.3030***	**0.2810***	**0.3058***	**0.2222***	**0.1743***	**0.2997***	**0.2010***	**0.0810***	**0.0452**	**0.1169**	**0.0544***

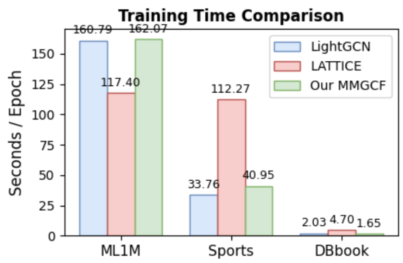

Figure 2: Comparison of the training time per epoch between MMGCF, LightGCN, and LATTICE.

Table 3: Sensitivity Analysis on ML1M

Parameter	Recall@10	NDCG@10	Recall@20	NDCG@20
GCN Layers				
1	0.1842	0.2991	0.2781	0.3027
2	0.1854	**0.3047**	0.2794	**0.3066**
3	**0.1871**	0.3030	**0.2810**	0.3058
Item Embedding Weight				
Alpha	**0.1871**	0.3030	**0.2810**	0.3058
Equal	0.1796	0.2848	0.2719	0.2905
Normalized	0.1854	**0.3047**	0.2794	**0.3066**
Fusion Strategy				
Concat	0.1769	0.2808	0.2700	0.2874
Mean	**0.1871**	**0.3030**	**0.2810**	**0.3058**
Sum	0.1664	0.2589	0.2544	0.2669

In contrast, MMGCN and GRCN underperform with respect to MMGCF despite substantially higher training costs. This highlights the capacity of MMGCF to extend LightGCN effectively, while maintaining scalability. On DBbook, a pure CF model (LightGCN) fails in providing accurate recommendations, likely due to the higher sparsity (Table 1). In this sparse context, multimodal models outperform LightGCN across all metrics. MMGCF achieves significant improvements over LATTICE, the best-performing baseline, while maintaining even lower training times. This result suggests that MMGCF effectively outperforms LightGCN in sparser scenarios by exploiting multimodal features, without compromising training time efficiency. The results on the Sports dataset further validate our previous findings. While LightGCN serves as a competitive baseline, MMGCF consistently achieves higher Recall and NDCG, showing that multimodal integration can significantly enhance recommendation accuracy over pure CF, while maintaining lower training times than most of the multimodal baselines. Notably, MMGCF is significantly faster than LATTICE. **To answer RQ1, we conclude that MMGCF successfully extend LightGCN through the strategic integration of multimodal features. The model consistently outperforms multimodal baselines, and demonstrates better efficiency, particularly in high sparsity scenarios.**

To address RQ2, we investigate MMGCF sensitivity across three key hyperparameters: GCN depth, fusion strategy, and embedding weighting. Table 3 reports the peak performance for each configuration, obtained by isolating one parameter and optimizing the others via grid search. Due to space contraints, we focus the discussion on ML1M, as findings were consistent across all datasets. First, we observe that the best results are obtained with a GCN depth of 3 layers. This aligns with the findings in the original LightGCN paper [7], which suggests that increased depth generally enhances performance. As for the weighting strategy, our results indicate that this component is critical for performance optimization. Both learnable *alpha* and normalized weighting strategies outperform equal weighting, the latter being the conventional approach in existing literature [6, 24, 25]. This confirms that treating all modalities as equally informative may lead to suboptimal performance. Instead, MMGCF effectively balances multimodal and CF signals, without adding noise. Finally, the choice of the fusion strategy significantly impacts performance. Mean fusion consistently outperforms both summation and concatenation, the latter being a common standard in the literature [6, 12, 31]. This suggests that balanced signal aggregation is more effective than simply increasing representation dimensionality. **To answer RQ2, our sensitivity analysis shows that embedding weighting and fusion strategy are critical. Specifically, assigning equal weights degrades performance, while normalizing embeddings or using a learnable parameter improves performance. Additionally, mean-based fusion generally outperforms the more common concatenation.**

5 Conclusions

In this paper, we presented MMGCF, a multimodal RS that integrates CF embeddings with pre-trained multimodal item features through efficient and effective weighting mechanisms and fusion strategies. Our experiments showed that MMGCF offers the optimal trade-off between recommendation accuracy and computational efficiency.

Future work will explore the generalizability of our weighting and fusion strategies across other recommendation models [6, 31], and evaluate how alternative CF backbone models influence MMGCF performance. Additionally, we intend to expand to include the environmental impact of these models by assessing the carbon footprint generated during training [3, 17].

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